

HydroSim

Learning Modules

Background reading on the growing systems, measurements and crops that HydroSim simulates.

Prepared for customers, industry partners, new team members, and thesis reviewers.

Every factual claim is cited.

Each module carries its own numbered Sources list, drawn from university cooperative extension publications, e-GRO grower factsheets, and peer-reviewed papers.

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HydroSim

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How to use this document

If you are	Read
New to hydroponics	Modules 1 to 4, in order
Growing a specific crop	Module 5 (lettuce) or 6 (tomato)
Working on the engine, or reviewing the model	Appendix A, then the module that covers the parameter
Looking up a term	The glossary at the end

Modules describe published horticultural practice. They do not describe how HydroSim works. Everything about the simulator's own numbers and limits is gathered in the appendices, so the two never get confused.

Quick reference

Published guidance for the two crops HydroSim supports.

Measurement	Lettuce	Tomato
Solution pH	5.5 to 6.0 [M4-3]	5.5 to 6.5 [M6-1], or 5.5 to 6.0 [M6-2]
EC	1.0 to 2.0 mS/cm [M4-3]	2.0 to 2.5 mS/cm [M6-1] [M6-2]
Nitrogen in solution	150 to 200 ppm [M4-3]	70 to 150 ppm [M6-1]
Dissolved oxygen	above 6 ppm [M3-1]	above 6 ppm [M3-1]
Usual growing system	NFT or DWC	Substrate: perlite, pine bark or rockwool [M6-1]
Signature disorder	Pythium root tip necrosis [M4-5]	Blossom end rot [M6-6]

Reference codes point to the numbered source in that module, so [M4-3] is source 3 in module 4. Air temperature and cycle length for hydroponic lettuce are not covered by the sources collected here. Module 5 gives field figures, which are a comparison rather than a target.

1. Nutrient Film Technique (NFT)

IN SHORT *A pump keeps a thin stream of nutrient solution running down sloped channels. Roots sit partly in the film and partly in air. It suits fast, shallow rooted crops such as lettuce, and it stops working the moment the pump does.*

How it works

NFT circulates a shallow stream of solution past the roots continuously. The stream is deliberately thin, so only the bottom of the root mat is wetted and the rest stays in air, giving the plant nutrients and oxygen at the same time. The method was first described by Dr. Allen Cooper at the Glasshouse Crops Research Institute in Littlehampton, England, in the 1960s.

Element	Detail
Closed loop	A pump lifts solution from a reservoir, it flows down the channels by gravity, and surplus is collected, replenished and reused rather than discarded.
Channels	Enclosed channels hold the plants. In the simplest builds this is a polyethylene tube with slits cut for the roots. Commercial channels are rigid extrusions with removable lids for cleaning.
Distribution	Solution reaches each channel through a manifold and flexible tubing joined by emitters. Shut-off valves let a grower tune flow rate per channel.
Tubing colour	White poly pipe reflects heat and keeps solution cooler. Black pipe does the opposite, which matters in warm climates.

Best suited to

Fast-growing, shallow-rooted crops. Lettuce, basil, cilantro, mint, parsley, spinach, arugula, kale, Swiss chard and mustard greens all do well, and some strawberry cultivars are grown this way in vertical or horizontal channels.

The system is efficient with both water and floor space, and its lightweight modular layout scales up or down without a redesign. Optimizing water and nutrient use is the goal the technique was designed around.

Watch out for

- Pump dependency. The film only exists while the pump runs. NFT holds no water reserve at the roots, so a pump or power failure starts stressing plants quickly rather than over hours.
- Clogs. Channels, emitters and delivery lines need regular checks to confirm the film is still flowing and has not been blocked by root mass or debris.
- Nutrient drift. The same solution recirculates, so concentration and balance shift as plants draw it down. This is ongoing maintenance, not a one-time setup concern.
- Heavy crops. Large, top-heavy or deep-rooted plants are a mismatch for shallow channels sized around leafy greens. See module 6 for what tomato needs instead.

Sources

1. Virginia Cooperative Extension (SPES-463): Hydroponic Production of Edible Crops, Nutrient Film Technique (NFT) Systems <https://www.pubs.ext.vt.edu/SPES/spes-463/spes-463.html>

2. Oregon State University Extension (EM 9457): Hydro hints, Nutrient film technique <https://extension.oregonstate.edu/catalog/pub/em-9457-hydro-hints-nutrient-film-technique>
3. Oklahoma State University Extension: Hydroponics (fact sheet) <https://extension.okstate.edu/fact-sheets/hydroponics>
4. Palmitessa, Signore & Santamaria (2024), Frontiers in Plant Science: Advancements and future perspectives in nutrient film technique hydroponic system <https://doi.org/10.3389/fpls.2024.1504792>

2. Deep Water Culture (DWC)

IN SHORT *Roots hang into a shallow pond of nutrient solution held up by a floating raft. There is no pump moving solution past the roots, so keeping the water oxygenated is the main job.*

How it works

DWC, also called a floating or raft system, suspends roots directly in aerated solution. Plants sit in net pots or soilless-media cells set into a raft that floats on the surface. In the 1980s, Dr. Merle Jensen at the University of Arizona's Environmental Research Lab developed a relatively deep raceway design, roughly 6 to 8 inches, that commercial growers and hobbyists went on to adopt for leafy greens.

Element	Detail
The pond	The production area is essentially a shallow pond of solution. There is no circulating film and no irrigation cycle to time.
Aeration	Water movement and air exchange are minimal, so DWC depends on active aeration, usually air pumps driving air stones, or ozone injection at larger scale.
Oxygen target	Above 6 ppm dissolved oxygen is optimal. Below that, growth can be inhibited and ethylene production increases.
Temperature limit	Warmer water holds less oxygen. At 35°C (95°F) and 5 dS/m, solubility tops out near 6.85 ppm no matter how hard you aerate.

Because of that last point, a chiller rather than a bigger air pump is the fix for a hot reservoir.

Monitoring and layout

- Meters. DWC needs meters for electrical conductivity, pH and dissolved oxygen. EC and pH should be checked two to three times a week. Growers running submerged-root systems specifically need to watch dissolved oxygen and water temperature.
- Level. Keep the reservoir bottom level to within about an inch so solution depth is even and growth does not vary across the bed.
- Drainage. A bulkhead fitting and valve at the bottom makes draining practical instead of a chore.

Watch out for

The open tank creates the main trade-offs. Light on an open surface encourages algae, and an exposed pond loses more water to evaporation than an enclosed system. Once salinity climbs too high, the whole nutrient bath has to be dumped and refreshed rather than topped up. At harvest, avoid letting solution drip from pulled roots onto neighbouring plants.

The upside is speed and simplicity. DWC accelerates growth, suits leafy greens and herbs, and has no distribution plumbing to clog.

Sources

1. Virginia Cooperative Extension (SPES-464): Hydroponic Production of Edible Crops, Deep Water Culture (DWC) Systems https://www.pubs.ext.vt.edu/content/pubs_ext_vt_edu/en/SPES/spes-464/spes-464.html

2. Oregon State University Extension (EM 9455): Hydro hints, Deep water culture <https://extension.oregonstate.edu/catalog/pub/em-9455-hydro-hints-deep-water-culture>
3. University of Missouri Extension (G6984): Hydroponic Nutrient Solutions <https://extension.missouri.edu/publications/g6984>
4. University of Kentucky Center for Crop Diversification (CCD-SP-20): Irrigation in Hydroponic Systems <https://ccd.uky.edu/sites/default/files/2024-12/ccd-sp-20-irrigation-in-hydroponic-systems.pdf>

3. Nutrient solution: pH, EC and dissolved oxygen

IN SHORT *These three readings decide whether a plant can use what you feed it. pH controls availability, EC reports total salt concentration, and dissolved oxygen is capped by water temperature.*

A nutrient solution is water plus fertilizers, which makes the water itself an ingredient. Testing your source water is the only way to know whether it is suitable at all.

pH: availability, not quantity

A well-mixed solution containing every element can still starve a plant if pH sits outside the workable range, because pH determines whether nutrients can be taken up.

Crop targets are narrower than general guidance suggests. Hydroponic lettuce solution is held between pH 5.5 and 6.0, while lettuce transplants raised in soilless substrate tolerate a wider 5.5 to 6.5 band. Tomato guidance is 5.5 to 6.5, or 5.5 to 6.0 depending on the source (module 6). For general EC and pH management practice, see Oklahoma State Extension's guide.

EC: total salts, not balance

EC reports the overall concentration of fertilizer salts, not which nutrients are present or in what ratio. Two solutions can share an EC reading and behave completely differently.

Lettuce grows best at roughly 150 to 200 ppm nitrogen, which corresponds to about 1.0 to 2.0 mS/cm. In deep water culture, EC and pH should be checked two to three times each week. Tracking EC over time is how a grower notices that total fertilizer in solution has shifted.

Dissolved oxygen and temperature

Above 6 ppm is optimal. Below it, growth may be inhibited and ethylene production rises. Water temperature caps how much oxygen the solution can hold, and warmer water holds less. At 35°C (95°F) with 5 dS/m EC, solubility is about 6.85 ppm, and additional aeration cannot push past it.

Chilling the solution raises that ceiling. Oxygen availability is a concrete example of how solution temperature ends up governing plant health and nutrient uptake.

Watch out for

Lighting, airflow, dissolved oxygen, humidity and air temperature can all push a crop outside its comfort zone even when solution chemistry checks out. When symptoms appear, check foliage and roots before adjusting fertilizer. The cause is frequently a root or water-management problem rather than the feed.

Sources

1. University of Missouri Extension (G6984): Hydroponic Nutrient Solutions <https://extension.missouri.edu/publications/g6984>
2. Oklahoma State University Extension: Electrical Conductivity and pH Guide for Hydroponics <https://extension.okstate.edu/fact-sheets/electrical-conductivity-and-ph-guide-for-hydroponics>
3. e-GRO Nutritional Monitoring Series (2018): Lettuce (*Lactuca sativa*) <https://hortamericas.com/wp-content/uploads/2018/04/e-gro-Nutritional-Factsheet-Lettuce.pdf>

4. Virginia Cooperative Extension (SPES-464): Hydroponic Production of Edible Crops, Deep Water Culture (DWC) Systems
https://www.pubs.ext.vt.edu/content/pubs_ext_vt_edu/en/SPES/spes-464/spes-464.html
5. University of New Hampshire Extension: Scouting & Managing Greenhouse Nutrient Problems (fact sheet)
<https://extension.unh.edu/resource/scouting-managing-greenhouse-nutrient-problems-fact-sheet>

4. Reading plant health

IN SHORT *Check crops weekly rather than waiting for symptoms to force a rescue. Look at foliage colour and vigour, look at the roots, and test pH and EC. Visual symptoms overlap, so they are a prompt to measure rather than a diagnosis.*

pH governs which nutrients are available, and EC reports the overall concentration of fertilizer salts.

Healthy signs

- Even, saturated leaf colour. Uniform green across the canopy, no pale cast on older leaves, no scorched margins.
- Steady size gain between checks. Growth rate is an early signal. In lettuce trials, reduced growth from nitrogen deficiency was visible within the first two weeks, before the yellowing was obvious.
- Turgid foliage. Plants that wilt while solution is plentiful are signalling a root-zone or salinity problem, not thirst.
- Intact roots. Foliar symptoms often trace back to the roots. A root system compromised by Pythium or waterlogging produces leaf symptoms that look nutritional but will not respond to a fertilizer change.

Warning signs

Sign	Usual meaning
Pale green foliage, overall stunting	Typical first presentation of nitrogen deficiency, which can progress to wilting and dead or yellow leaf margins. In lettuce it advances to uniform chlorosis of the older leaves.
Wilting despite adequate solution, dark green leaves, burned margins	Excess soluble salts, a chemically induced drought that can progress to root death.
Lower-leaf chlorosis, or purpling on red cultivars	In lettuce, associated with low fertility (low EC) rather than disease.
Calcium disorders	Often not a fertilizer shortfall. Under-fertilization, nutrient imbalance and low pH contribute, and so do high temperature, low humidity and poor airflow, because calcium moves with the transpiration stream and leaves out-compete fruit for it.
Root necrosis	Symptoms differ by crop. Spinach shows severe rot through the whole root system, while Pythium on lettuce presents as root tip necrosis.

The calcium row has a specific and well documented expression in tomato, blossom end rot. Module 6 covers it, because both the mechanism and the management are tomato specific.

How to diagnose

Nutrient solution temperature, EC, dissolved oxygen and pH all shape how root disease presents and progresses, so an environmental reading is part of the diagnosis rather than background context.

For reference points on a common crop: hydroponic lettuce grows best around 150 to 200 ppm nitrogen, roughly 1.0 to 2.0 mS/cm, with solution pH held between 5.5 and 6.0.

Sources

1. University of New Hampshire Extension: Scouting & Managing Greenhouse Nutrient Problems (fact sheet) <https://extension.unh.edu/resource/scouting-managing-greenhouse-nutrient-problems-fact-sheet>
2. Penn State Extension: Hydroponics Systems and Principles of Plant Nutrition, Essential Nutrients, Function, Deficiency, and Excess <https://extension.psu.edu/hydroponics-systems-and-principles-of-plant-nutrition-essential-nutrients-function-deficiency-and-excess>
3. Mattson, e-GRO Research Update (2015): Symptoms of Common Nutrient Deficiencies in Hydroponic Lettuce https://www.e-gro.org/pdf/Mattson_Lettuce_2015_9.pdf
4. e-GRO Nutritional Monitoring Series (2018): Lettuce (Lactuca sativa) <https://hortamericas.com/wp-content/uploads/2018/04/e-gro-Nutritional-Factsheet-Lettuce.pdf>
5. Plant Health Progress (APS), 2024: Diagnostic Guide for Pythium Root Rot in Hydroponic Leafy Green and Herb Production <https://doi.org/10.1094/PHP-07-24-0070-DG>

5. Lettuce

IN SHORT A fast leafy crop that suits both NFT and DWC. It is harvested during heading, before it flowers, so it has no fruiting phase. Solution targets are pH 5.5 to 6.0 and 1.0 to 2.0 mS/cm.

Growing system

Lettuce is the standard crop for both systems in this document. It is named first among the crops suited to NFT, and DWC is widely used for leafy greens.

Environment targets

Hydroponic lettuce grows best around 150 to 200 ppm nitrogen, or roughly 1.0 to 2.0 mS/cm, with solution pH between 5.5 and 6.0. Transplants raised in soilless substrate tolerate a wider pH band of 5.5 to 6.5.

Growth stages

The clearest published stage sequence for lettuce comes from field production. The University of Arizona describes six distinct stages: seed, cotyledon, seedling, rosette, cupping and heading.

Stage	Begins when	Fall planting	Winter planting
Seed	Pre-planting to emergence. Germination starts once the seed meets water at a suitable temperature	as little as 12 hours	up to 7 days
Cotyledon	Plant sheds the seed coat and emerges. Lasts until roots are better established	n/a	n/a
Seedling	First true leaf emerges, once the root has grown a couple of inches	about 7 days from emergence	about 20 days
Rosette	A distinct circular cluster of leaves has formed	about 25 days	up to 50 days
Cupping	Inner leaf tips curl inward at the edges, signalling that head formation is near	about 7 days	about 14 days
Heading	Cupped leaves overlap and cover the growing point. Runs to harvest	about 30 days	about 45 days

All figures from the University of Arizona source. Fall-planted lettuce may need as little as 65 days from the start of germination to harvest, while winter-planted lettuce can take as long as 120 days. The difference is driven by temperature.

Two caveats. First, these figures are from field production in the Arizona low desert, where soil, irrigation and open-sky conditions differ from a greenhouse. The stage names and their sequence transfer. The durations are a comparison point, not a target. Second, lettuce never reaches a flowering or fruiting stage in production, because it is cut during heading.

Watch out for

Pythium root rot presents on lettuce as root tip necrosis rather than the whole root system rotting. Low fertility shows as lower-leaf chlorosis on green cultivars and dark purpling on red ones. Module 4 covers both in context.

Sources

1. University of Arizona, Arizona Pest Management Center (ACIS): Lettuce, Crop Management. Kerns, Matheron, Palumbo, Sanchez, Still, Tickes, Umeda & Wilcox <https://acis.cals.arizona.edu/agricultural-ipm/vegetables/lettuce/crop-management>
2. Oregon State University Extension (EM 9457): Hydro hints, Nutrient film technique <https://extension.oregonstate.edu/catalog/pub/em-9457-hydro-hints-nutrient-film-technique>
3. Oregon State University Extension (EM 9455): Hydro hints, Deep water culture <https://extension.oregonstate.edu/catalog/pub/em-9455-hydro-hints-deep-water-culture>
4. e-GRO Nutritional Monitoring Series (2018): Lettuce (Lactuca sativa) <https://hortamericas.com/wp-content/uploads/2018/04/e-gro-Nutritional-Factsheet-Lettuce.pdf>
5. Plant Health Progress (APS), 2024: Diagnostic Guide for Pythium Root Rot in Hydroponic Leafy Green and Herb Production <https://doi.org/10.1094/PHP-07-24-0070-DG>

6. Tomato

IN SHORT *A long-season fruiting crop that behaves very differently from lettuce. It runs for months, needs pollinating, and is normally grown in substrate rather than in NFT or DWC. Solution targets are pH 5.5 to 6.5 and 2.0 to 2.5 mS/cm.*

Tomato is the most widely produced greenhouse crop in the world, with a large share of fresh-market supply coming from Canada and Mexico. Greenhouse growers use indeterminate varieties, which keep growing after setting fruit, and train them vertically on a trellis. Field operations use determinate varieties, which stop growing once fruiting has occurred.

Growing system

This is where tomato diverges from everything above. Substrate-based systems are normally used, because large vining plants need more root support than traditional hydroponic systems can provide.

Substrate	Used in
Perlite	Bag culture, Dutch buckets (11-litre buckets with a drainage hole)
Pine bark	Container systems, 3 or 5 gallon nursery pots
Rockwool	Highwire systems in advanced greenhouses in cooler regions, with trellis wires at least 12 feet up rather than the usual 6 to 8

Purdue's controlled environment programme likewise describes the Dutch bucket system for greenhouse tomato.

HydroSim offers NFT and DWC only. A tomato grown in either setting inside the simulator is a simplification, not a description of standard commercial practice. Appendix B lists this alongside the other boundaries.

Environment targets

Variable	Published guidance
Daytime temperature	27 to 29°C (80 to 85°F) for significant growth and yield. Purdue gives a working band above 18°C (65°F) and below 29°C (85°F)
Night temperature	16 to 18°C (60 to 65°F), but no lower
EC	2.0 to 2.5 mS/cm. Purdue gives roughly the same, noting higher EC improves flavour
pH	5.5 to 6.5. Purdue gives 5.5 to 6.0
CO ₂	Enrichment up to 600 ppm
Light	Above 20 mol/m ² /day
Humidity	60 to 70 percent is where pollination works best

Cold damage shows as leaves turning purple and curling inward. Excessive heat affects fruit development, causing cracked fruit and reduced fruit set.

Growth stages and crop cycle

Transplants are usually produced in 4 to 6 weeks and moved on at 4 to 6 inches tall. Purdue puts flowering at 6 to 8 weeks after planting, with harvest running 2 to 4 times each week for 15 to 20 weeks. Alabama Extension describes harvesting 1 to 3 times a week depending on season.

Two scheduling patterns are common. One is a single 10-month crop started in August and finished the following June. The other is the two-crop method developed by Mississippi State University, which also starts in August, ends the first crop in December, and immediately transplants a second for spring.

For stage terminology, the BBCH scale codes development into ten principal stages numbered 0 to 9: germination, leaf development, formation of side shoots, stem elongation, inflorescence emergence, flowering, fruit development, ripening and senescence. It is the standard vocabulary if stage names need to be recognisable to a horticulturist.

Pollination

Tomato flowers are perfect, meaning they carry male and female parts and can self-pollinate, but cross-pollination gives better fruit set and yield. Elevated temperature and high humidity both reduce pollination, so the recommended window is the middle of the day before it gets too hot, with humidity between 60 and 70 percent.

Growers use bumblebees, mechanical pollinators, or electric pollinators that vibrate the flower truss. Hives stay active 6 to 8 weeks. Pollination is done three times a week as new flowers develop.

Watch out for: blossom end rot

The characteristic tomato disorder, and the one most closely tied to the variables a simulator tracks. It is physiological rather than infectious, and is not caused by fungi or bacteria. It appears as a light tan, water-soaked blemish at the blossom end that enlarges into a dark, sunken, leathery lesion, and it can also occur on the side of the fruit or internally with nothing visible outside.

The cause is a localised calcium deficiency in the developing fruit rather than a shortage of calcium in the solution. Contributing factors include low available calcium, excess nitrogen, ammoniacal nitrogen, high soluble potassium and magnesium, high salinity, and both inadequate and excess moisture. Root damage from disease or heavy pruning raises the risk. In greenhouse systems, not cycling irrigation at night increases blossom end rot,

because night is an important period for calcium uptake. Oregon State adds that calcium and water demand rise as temperature and growth rate rise, so heat spells favour the disorder.

Sources disagree on humidity. New Hampshire Extension reports that many studies find increasing humidity reduces blossom end rot. University of Florida guidance lists low humidity among the causes. The mechanism in both cases is competition between leaves and fruit for calcium moving in the transpiration stream, but the practical direction is contested. Treat any humidity coefficient here as unsettled.

Foliar calcium sprays are not effective, because very little calcium is taken up by the fruit and calcium absorbed by leaves cannot be moved to it. Commercial operations inject two separate stock solutions, a general fertilizer and a calcium nitrate solution, kept apart so calcium phosphate does not precipitate.

Nutrient solution concentration ranges

From Alabama Extension, adapted from Mills and Jones (1996) and Hochmuth and Hochmuth (2018).

Nutrient	ppm	Nutrient	ppm
Nitrogen	70 to 150	Iron	2.8
Phosphorus	50	Manganese	0.8
Potassium	120 to 200	Boron	0.7
Calcium	150	Copper	0.2
Magnesium	40 to 50	Zinc	0.3
Sulfur	50 to 60	Molybdenum	0.05

Sources

1. Alabama Cooperative Extension System (ANR-2955): Greenhouse Tomato Production. Blanchard, Pickens, da Silva & Wells, reviewed February 2025 <https://www.aces.edu/blog/topics/crop-production/greenhouse-tomato-production/>
2. Purdue University, Controlled Environment Agriculture: Greenhouse Tomato Production <https://www.purdue.edu/hla/sites/cea/wp-content/uploads/sites/15/2024/05/Greenhouse-Tomato-Production.pdf>
3. Meier (ed.), BBCH Monograph: Growth Stages of Mono- and Dicotyledonous Plants. Also cited in docs/DATASET.md https://www.openagrar.de/servlets/MCRFileNodeServlet/openagrar_derivate_00010428/BBCH-Skala_en.pdf
4. University of New Hampshire Extension: Growing Vegetables, Managing Blossom End-Rot (fact sheet) <https://extension.unh.edu/resource/growing-vegetables-managing-blossom-end-rot-fact-sheet-0>
5. Oregon State University Extension (FS 139): Blossom-End Rot of Tomatoes <https://extension.oregonstate.edu/catalog/fs-139-blossom-end-rot-tomatoes>
6. University of Florida IFAS, u-scout: Blossom End Rot (tomato) <https://plantpath.ifas.ufl.edu/u-scout/tomato/blossom-end-rot.html>
7. HortScience 47(12), 2012: Changes in Selected Quality Attributes of Greenhouse Tomato Fruit as Affected by Pre- and Postharvest Environmental Conditions in Year-round Production <https://journals.ashs.org/view/journals/hortsci/47/12/article-p1698.xml>

Appendix A: how these numbers relate to the engine

For the development team, reviewers and technically minded partners. Customers using the product do not need this section.

HydroSim's growth and stress model is a grey-box: a transparent agronomy-based model with hand-set parameters that are then calibrated. Some of those parameters come from the published guidance in the modules above. Others are engineering estimates. This appendix separates the two, so nothing in the codebase looks better sourced than it is. docs/engine.md explains the formulae themselves.

Where the engine agrees with the sources

- Lettuce solution pH 5.5 to 6.0 and EC in the 1.0 to 2.0 range sit inside published guidance (module 3).
- Tomato stage names, seedling through fruiting to harvest, are broadly consistent with the BBCH principal stages (module 6).
- DWC being more forgiving than NFT is directionally sound. DWC holds a large buffered reservoir (module 2) while NFT holds no reserve at the roots (module 1).

Where the engine differs from the sources

The tomato profile in `backend/app/sim/engine.py` is labelled `purdue_greenhouse_tomato_profile`. Several values do not match what Purdue and Alabama publish. None are necessarily wrong, but they should be recorded as deliberate choices.

Parameter	Engine	Published guidance	Note
Tomato EC, crop level	2.5	2.0 to 2.5 [M6-1] [M6-2]	top of the published band
Tomato EC, fruiting and harvest	3.5	2.0 to 2.5 [M6-1] [M6-2]	above both sources
Tomato CO ₂ , vegetative and flowering	1000 ppm	enrichment up to 600 ppm [M6-2]	well above the cited figure
Tomato air temperature	25°C (77°F)	27 to 29°C daytime [M6-1]	below the Alabama daytime band
Night temperature	not modelled	16 to 18°C [M6-1]	one air temperature, no day and night split
Tomato cycle length	120 days	about 10 months for a single crop [M6-1]	much shorter than a commercial greenhouse crop
System choice for tomato	NFT or DWC	substrate culture [M6-1]	see module 6

The EC ramp is the most defensible of these. Purdue notes higher EC improves flavour, and published work shows that raising solution EC in a rockwool highwire system increased lycopene and total soluble solids. The direction has a real basis. The specific value of 3.5 does not come from any source collected here.

Parameters with no source

These are hand-set and currently uncited. They are listed so that a partner or reviewer can challenge them directly, and so that calibration work has a worklist.

Constant in <code>engine.py</code>	What it controls
STRESS_WEIGHTS	Relative impact of each variable on the stress score
TOLERANCES	How far from target a variable can drift before it saturates its stress contribution
LIMITS	Hard plausible bounds per variable
SYSTEM_TOLERANCE_FACTORS	NFT 1.0 against DWC 1.3. The direction is sound, the 1.3 is an estimate, as the code comment already notes
stage_targets	Per-stage optima for both crops
Cycle lengths	Lettuce 45 days, tomato 120 days

Reference codes point to the numbered source in that module, so [M6-2] is source 2 in module 6.

Appendix B: what the simulator does not model

Stated plainly so that anyone relying on HydroSim knows the boundaries of what it can tell them.

- Pest and disease pressure. Pythium and blossom end rot are described in these modules but are not simulated.
- Individual nutrients. EC is an aggregate, so the engine cannot represent a nitrogen shortfall alongside a calcium excess.
- Substrate culture. Only NFT and DWC are offered, though greenhouse tomato is normally grown in perlite, pine bark or rockwool.
- Pollination and fruit set for tomato.
- Day and night temperature separately. There is one air temperature value.
- Light spectrum and daily light integral.
- Cultivar differences and root zone volume.

Glossary

Term	Meaning
BBCH	A standard numeric scale for coding plant growth stages, 0 to 9
Bolting	Running to flower, which ends a leafy crop's marketable life
Chlorosis	Yellowing of leaf tissue
Determinate	A tomato variety that stops growing once fruiting has occurred. Field practice
DO	Dissolved oxygen, measured in ppm
Dutch bucket	An 11-litre container with a drainage hole, used with perlite for greenhouse tomato
DWC	Deep Water Culture. Roots suspended in an aerated pond of solution
EC	Electrical conductivity, a measure of total dissolved salts, in mS/cm or dS/m
Indeterminate	A tomato variety that keeps growing after setting fruit. Greenhouse practice
Necrosis	Death of plant tissue, usually appearing brown or black
NFT	Nutrient Film Technique. A thin recirculating stream of solution in sloped channels
Pythium	A water mould causing root rot in hydroponic systems
Rockwool	A spun mineral substrate used in highwire tomato systems
Rosette	The lettuce stage where leaves form a distinct circular cluster
Truss	A cluster of flowers or fruit on a tomato stem

Related documents

- docs/DATASET.md: the reference ranges used to generate the synthetic dataset.

- docs/engine.md: how those ranges feed the grey-box stress and growth model.